Latency Numbers & Powers of Two

the handful of numbers to know by heart

Watch the 3-min video

In one line: know powers of two (KB, MB, GB, TB, PB), seconds per day (86,400 ≈ 10^5), and the latency ladder from 1 ns (L1) to 150 ms (across continents). With them you turn "1 million requests a day" into "about 12 per second" in two seconds.

1 Why memorise a few numbers

An interviewer says: "we get one million requests a day. Is that a lot?" Most people reach for a calculator. An experienced engineer answers in two seconds: **about twelve requests per second**. A single server handles that easily. The trick is a handful of numbers you know by heart: powers of two, seconds in a day, and latency orders of magnitude.

2 Powers of two: data sizes

2^{10} = 1,024, roughly a thousand. That is a **kilobyte**. 2^{20} is about a million, a **megabyte**. 2^{30} is about a billion, a **gigabyte**. Then a **terabyte** (about a trillion bytes) and a **petabyte** (about a thousand trillion). Each step is a thousand times bigger.

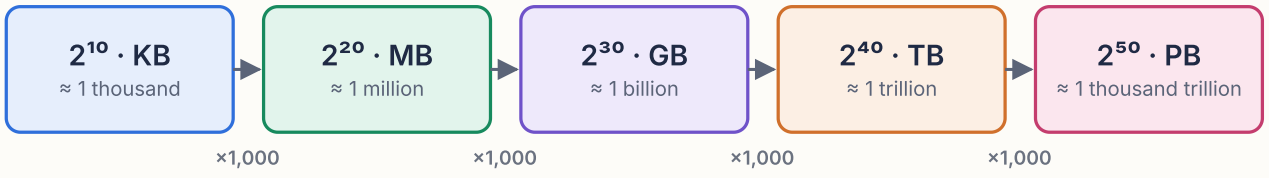


Figure 1. The powers-of-two ladder. For estimates, treat 2^{10} as 1,000; the 2.4% difference per step does not change any design decision.

3 Time: seconds in a day, month and year

A day has **86,400 seconds**. Round it to **100,000** (10^5). So one million per day is about 10 per second, and more precisely about 12. A month is about **2.5 million** seconds and a year about **30 million**.

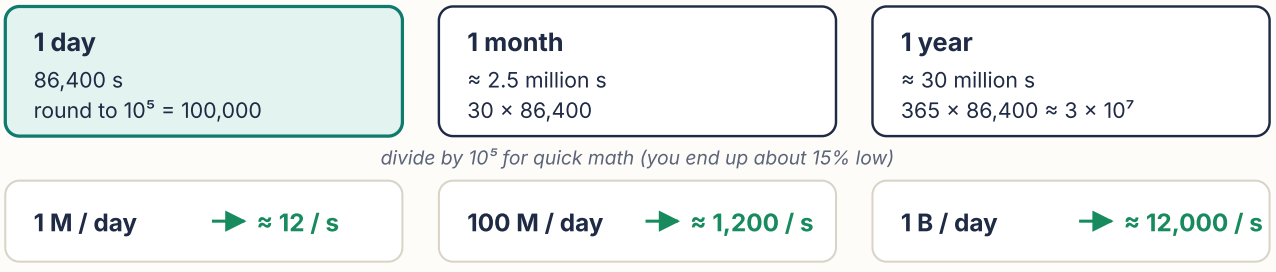


Figure 2. The time constants, and the three per-day → per-second conversions worth knowing cold.

4 The latency ladder

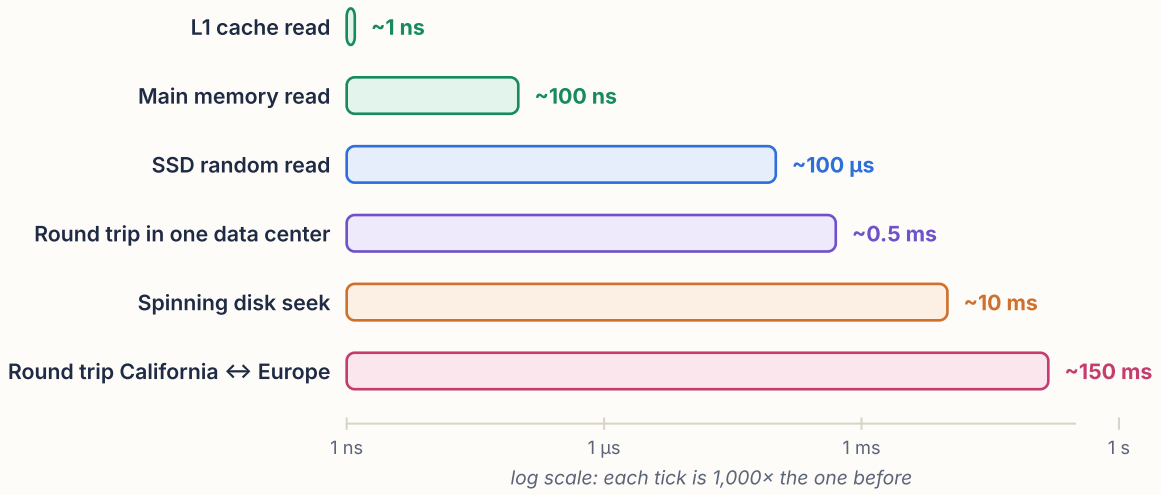


Figure 3. Orders of magnitude, not exact figures: hardware varies 2–3×, but the gaps between rungs stay huge.

Operation	Latency	If a memory read took 1 second...
L1 cache read	~1 ns	0.01 s
Main memory read	~100 ns	**1 s**
SSD random read	~100 µs	~17 minutes
Round trip in one data center	~0.5 ms	~1.4 hours
Spinning disk seek	~10 ms	~28 hours: more than a day
Round trip California ↔ Europe	~150 ms	~17 days

How much slower is a disk seek than a memory read? 10 ms ÷ 100 ns = $10^{-2} \div 10^{-7}$ = **100,000×**. That is why caches, indexes and sequential reads matter so much.

5 How the numbers drive design

- **Memory is fast but limited**, so cache the hot data in RAM.
- **A network hop inside a data center (~0.5 ms) is cheap** compared with a disk seek (~10 ms). That is why a remote cache like Redis is still fast.
- **Cross-continent calls are slow (~150 ms)**, and no amount of hardware fixes the speed of light. Serve users from a nearby region, or a CDN.
- **Avoid many sequential calls in one request.** Their latencies add up: five calls across continents, one after another, cost about 750 ms before any work is done.

WORKED EXAMPLE: 100 MILLION REQUESTS A DAY

- **Average RPS:** 100 M ÷ 10^5 s ≈ **1,000 requests/s** (exactly: ~1,157).
- **Peak:** a few thousand per second, at a 2–3× peak factor.
- **Bandwidth:** 1,000 × 10 KB = **10 MB/s** on average, about 80 Mbit/s. A single 10 Gbit/s network card has plenty of room.

COMMON MISTAKES

- Mixing up units: ms vs µs vs ns, or MB (bytes) vs Mb (bits, 8× smaller).
- Dividing by 86,400 with a calculator in the interview instead of using 10^5.
- Treating a network call as "slow" by default: inside one data center it is faster than a disk seek.
- Chaining calls across regions in one request and expecting low latency.

6 Key terms

Order of magnitude: the power of ten: 10 ms vs 100 ms

Round trip (RTT): time for a message to go and come back

KB / MB / GB / TB / PB: ≈ 10^3 / 10^6 / 10^9 / 10^{12} / 10^{15} bytes

Seek: moving a disk head to the data

ns / µs / ms: 10^{-9} / 10^{-6} / 10^{-3} seconds

Sequential read: reading data stored next to each other

7 Interview questions

Q1. How many seconds are in a day? BASIC

86,400. Round to 10^5 for quick math, so 1 million requests a day is about 10 per second, and more precisely about 12.

Q2. How much slower is a disk seek than a memory read? BASIC

About 10 ms versus 100 ns: roughly 100,000× slower. If a memory read took one second, a disk seek would take more than a day.

Q3. What is 2^{30} bytes, roughly? BASIC

About a billion bytes: a gigabyte. 2^{10} ≈ thousand (KB), 2^{20} ≈ million (MB), 2^{40} ≈ trillion (TB).

Q4. 100 million requests a day, 10 KB each. Estimate RPS and bandwidth. SCENARIO

100 M ÷ 10^5 ≈ 1,000 RPS on average, a few thousand at peak. 1,000 × 10 KB = 10 MB/s (about 80 Mbit/s) on average; size for the peak.

Q5. Your service in Virginia calls a database in Frankfurt four times per request. What is the problem? SCENARIO

Each round trip across the Atlantic is roughly 80–100 ms, so four sequential calls cost 300–400 ms before any work. Move the data or service into the same region, batch the calls, or cache.

Q6. Why is a Redis cache over the network still fast? SCENARIO

A round trip inside one data center is about 0.5 ms and Redis serves from RAM. That is about 20× faster than one spinning-disk seek (~10 ms) and much faster than a database query that touches disk.

Q7. Is a round trip inside a data center faster than an SSD random read? TRICKY

No. An SSD random read is about 100 µs; a data-center round trip is about 500 µs, so the SSD is roughly 5× faster. Both are far faster than a spinning-disk seek or a cross-continent call.

Q8. Why do sequential reads matter so much on a spinning disk? TRICKY

Each seek costs ~10 ms, then data streams quickly. Reading 1 MB in one run costs one seek; reading it as 256 random 4 KB blocks costs 256 seeks, over 2.5 seconds. That is why databases favour logs and large pages.

Next: SD-D-003 · How to Estimate RPS backendroadmap.com